

Temporal and Behavioral Patterns of Toxicity and Sentiment in Twitter

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Abstract—This study investigates two hypotheses about toxicity in social media: (H1) toxicity varies with the day of week and hour, and (H2) toxicity is equivalent to negative sentiment (i.e., toxic tweets are always negative). We use two datasets: Sentiment140 (1.6M tweets) and Twitter Toxic Tweets (28K after cleaning). A scalable PySpark pipeline is built: tokenization, CountVectorizer, IDF, and a weighted Naive Bayes classifier. The optimal probability threshold is 0.55, giving recall = 0.695 and F1 = 0.502 for the toxic class. Applying the model to all Sentiment140 tweets and aggregating by time and user, we find that H1 is confirmed (toxicity peaks on Tuesdays and at 21:00, lowest on Saturdays and early morning; ANOVA $p < 0.001$). H2 is rejected: the correlation between average user toxicity and positive sentiment ratio is below 0.04, and the difference between sentiment groups is less than 0.01. Our code is available on GitHub.

I. INTRODUCTION

Online toxicity has become a major concern on social media platforms. Understanding when users post toxic content and whether toxicity always aligns with negative sentiment is crucial for monitoring and protecting vulnerable audiences. This work addresses two precise hypotheses:

- **H1:** The level of toxicity depends on the day of the week and the hour of the day.
- **H2:** Toxicity is equivalent to negative sentiment, i.e., toxic tweets are always negative.

To test these hypotheses at scale, we employ two publicly available Twitter datasets, perform text cleaning and feature engineering using PySpark, train a toxicity classifier, and apply it to 1.6 million tweets. We evaluate statistical significance using ANOVA, Tukey’s HSD, Pearson/Spearman correlations, and Kruskal–Wallis tests. The complete reproducible code is provided in a GitHub repository [3].

II. DATASETS AND PREPROCESSING

A. Sentiment140

Sentiment140 [1] contains 1.6 million tweets with columns: text, date, polarity (0 = negative, 4 = positive), and user. Preprocessing steps:

- Text cleaning: lowercasing, removal of URLs, mentions, special characters, and extra spaces.
- Deduplication based on (user, date, cleaned_text).

- Extraction of weekday (0 = Monday, ..., 6 = Sunday) and hour from the timestamp.
- Mapping of polarity to binary form ($0 \rightarrow 0$, $4 \rightarrow 1$) for computing the ratio of positive tweets.

After cleaning, we obtain 1.59 million unique records (Figure 1a shows the perfect balance of the original polarity labels).

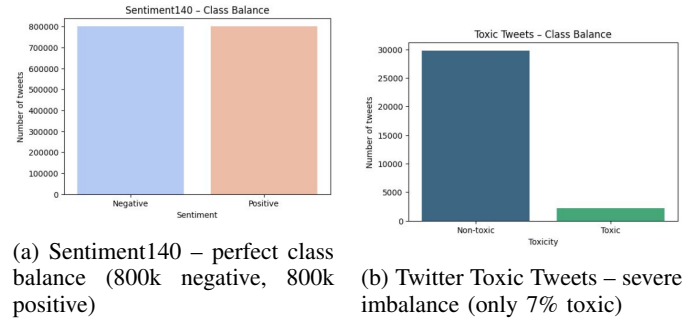


Fig. 1: Class distribution of the two datasets.

B. Twitter Toxic Tweets

The Twitter Toxic Tweets dataset [2] originally contains 32,000 tweets with a binary toxicity label (0 = non-toxic, 1 = toxic). The dataset is highly imbalanced (only 7% toxic). Cleaning is identical to Sentiment140, and duplicates are removed based on `cleaned_text`, resulting in 28,000 examples. The extreme class imbalance is visualised in Figure 1b.

III. TOXICITY CLASSIFIER

A. Pipeline and Model Selection

All experiments share a common base pipeline: Tokenizer $\rightarrow$ Feature extraction $\rightarrow$ IDF $\rightarrow$ Classifier. Inverse Document Frequency (IDF) is always applied to reduce the influence of very frequent words.

For feature extraction we compare two methods:

- HashingTF (numFeatures = 10000) – fast, no vocabulary stored.
- CountVectorizer (vocabSize = 10000) – builds an explicit vocabulary.

Three classifiers are evaluated:

- Logistic Regression with class weights (weighted LR).

- Random Forest (50 trees, maxDepth = 10) with class weights.
- Naive Bayes with class weights.

All models are trained on the same 80/20 split of the cleaned Toxic Tweets dataset (22,779 training, 5,662 test). To handle the severe class imbalance (only 7% toxic), instance weights are set as:

$$w_{\text{toxic}} = \frac{N_{\text{train}}}{2 \cdot N_{\text{toxic}}}, \quad w_{\text{non-toxic}} = \frac{N_{\text{train}}}{2 \cdot N_{\text{non-toxic}}}$$

which gives $w_{\text{toxic}} \approx 7.6$ and $w_{\text{non-toxic}} \approx 0.54$.

B. Threshold Tuning and Results

For each model we tune the probability threshold from 0.3 to 0.7 in steps of 0.05, selecting the value that maximises the F1 score for the toxic class. Table I reports the best results for each configuration at their optimal thresholds.

TABLE I: Performance comparison (toxic class) – optimal thresholds

Feature extraction	Classifier	Threshold	Precision	Recall	F1
HashingTF + IDF	Logistic Regression (weighted)	0.50	0.3606	0.5531	0.4366
HashingTF + IDF	Random Forest	0.50	0.2660	0.7003	0.3856
HashingTF + IDF	Naive Bayes	0.50	0.3672	0.5804	0.4498
CountVectorizer + IDF	Naive Bayes	0.55	0.3935	0.6948	0.5025

The best trade-off between precision and recall is achieved by **Naive Bayes with CountVectorizer + IDF** at threshold 0.55. Although its precision is modest (0.39), the recall of 0.695 is sufficiently high to capture temporal fluctuations in toxicity. False positives are acceptable for trend analysis because they are unlikely to distort relative differences across time.

Note: Adding bigrams (n=2) to the CountVectorizer pipeline decreased performance significantly ($F1 < 0.30$); therefore we retain the unigram model.

IV. APPLYING THE MODEL AND FEATURE ENGINEERING

The trained model is applied to all 1.59 million Senti-ment140 tweets, producing a `toxicity_score` (probability) and a binary `is_toxic` label using threshold 0.55.

A. User-level Aggregation

For each user we compute:

- `tweet_count`: number of tweets.
- `avg_toxicity`: mean of `toxicity_score`.
- `toxic_ratio`: proportion of tweets with `is_toxic=1`.
- `avg_sentiment_pos_ratio`: mean of the binarised polarity (0/1).

B. Time-level Aggregation

For each combination of weekday (0–6) and hour (0–23) we compute:

- `total_tweets`: number of tweets published in that specific weekday–hour slot.
- `avg_toxicity`: average `toxicity_score` of tweets in the slot, i.e. the mean predicted probability of toxicity.

- `toxic_ratio`: proportion of tweets in the slot that were classified as toxic (`is_toxic = 1`). This is the main metric for evaluating temporal patterns of toxicity.
- `avg_sentiment_pos_ratio`: mean of the binarised polarity (0/1) for tweets in the slot, representing the fraction of positive tweets.

V. HYPOTHESIS TESTING

A. Hypothesis 1: Temporal Dependence

We use the aggregated `toxic_ratio` across weekdays and across hours. One-way ANOVA reveals significant differences for both factors:

- Weekday: $F = 7.16$, $p \approx 0.0000$
- Hour: $F = 2.78$, $p = 0.0001$

Tukey’s HSD post-hoc test shows that Tuesday differs significantly from Saturday and Sunday, and that 21:00 differs from 6, 8, and 9 in the morning. Figure 2a illustrates that toxicity peaks on Tuesday and is lowest on Saturday; Figure 2b shows a peak at 21:00 and a minimum in the early morning. The joint effect of weekday and hour is visualised in the Figure 2c. Thus, **H1 is confirmed**.

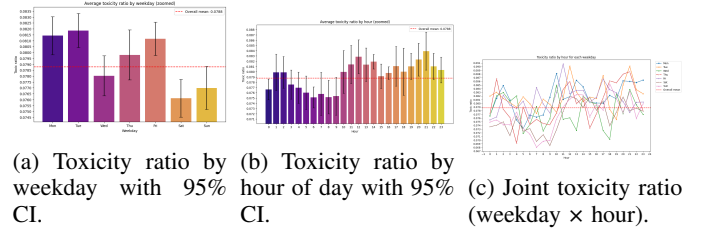


Fig. 2: Temporal patterns of toxicity.

B. Hypothesis 2: Toxicity vs. Sentiment

We test H2 (toxicity always equals negative sentiment) at the user level using:

- Pearson and Spearman correlations between `avg_sentiment_pos_ratio` and `avg_toxicity`.
- Splitting users into four sentiment groups:
 - Q1: always negative ($ratio = 0$)
 - Q2: mostly negative ($0 < ratio < 0.5$)
 - Q3: mostly positive ($0.5 \leq ratio < 1$)
 - Q4: always positive ($ratio = 1$)

For each group we compute mean toxicity and 95% confidence intervals.

Results:

- Pearson $r = 0.0187$ ($p = 3.09 \times 10^{-52}$), Spearman $r = 0.0386$ ($p = 4.11 \times 10^{-216}$). Both correlations are extremely weak (see Figure 3a).
- Kruskal–Wallis test gives $H = 56981$, $p \approx 0$ – statistical significance, but the actual differences are minuscule (Figure 3b):
 - Q1 (always negative): mean toxicity = 0.07658
 - Q4 (always positive): mean toxicity = 0.08548
 - Difference < 0.009 .

Therefore, **H2 is rejected**. Toxicity is not equivalent to negative sentiment; it can occur in tweets with any sentiment.

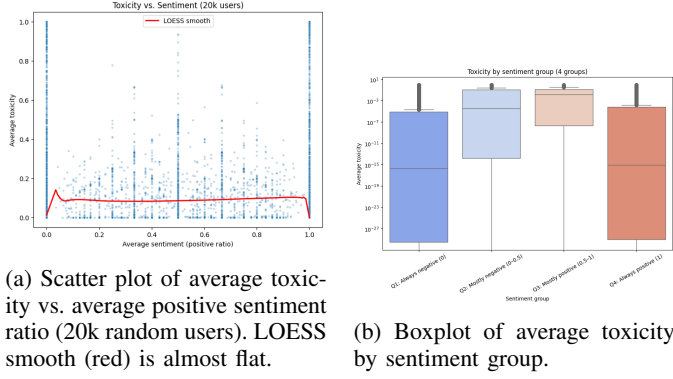


Fig. 3: Toxicity versus sentiment analysis.

VI. DISCUSSION

The temporal patterns found in this study are statistically robust, although the effect sizes are moderate. For instance, the difference in toxic ratio between Tuesday (the peak) and Saturday (the trough) is approximately 0.005, i.e., 0.5 percentage points. While such an absolute difference may appear small, from a big-data perspective it is highly meaningful: a relative increase of about 6–7% in toxicity on Tuesday compared to the weekly average can be exploited for real-time monitoring and early warning systems. Detecting these regular fluctuations allows platform moderators to allocate resources more efficiently (e.g., increasing scrutiny on Tuesday evenings) and helps researchers understand the social rhythms of online aggression.

The lack of correlation between toxicity and sentiment confirms that these two constructs measure fundamentally different aspects of online behavior. A tweet can be sarcastic or aggressively ironic while still having a positive surface sentiment – for example, “*I absolutely love this show! Anyone who disagrees is an idiot.*” Here the sentiment is positive, but the statement is clearly toxic. Conversely, a tweet expressing sadness or anger (“*I feel terrible today*”) may be negative in sentiment but not toxic at all. Thus, sentiment analysis alone cannot replace toxicity detection; both are necessary for a complete understanding of online discourse.

A. Limitations:

- Sentiment140 was collected in 2009; language and toxicity norms may have evolved.
- Precision of the classifier is low (0.39), making it unsuitable for automatic moderation but adequate for trend analysis.
- Network effects (retweets, replies) are not considered.

B. Reproducibility:

All code (EDA, training, labeling, analysis) is available at GitHub [3].

VII. CONCLUSION

We have tested two hypotheses on large Twitter data using a scalable PySpark pipeline. Hypothesis 1 (toxicity depends on weekday and hour) is confirmed: toxicity peaks on Tuesdays and at 21:00, and is lowest on Saturdays and in the early morning hours. Hypothesis 2 (toxicity equals negative sentiment) is rejected: the correlation between toxicity and sentiment is extremely weak (below 0.04), indicating that toxic tweets can appear with any sentiment. The trained classifier achieves a recall of 69.5%, which is sufficient for detecting temporal trends, even though precision is moderate. Our open-source implementation enables further research on the temporal dynamics of online toxicity.

REFERENCES

- [1] M. Michailidis, A. Go, R. Bhayani, and L. Huang, “Sentiment140 dataset with 1.6 million tweets,” Kaggle, 2009. [Online]. Available: <https://www.kaggle.com/datasets/kazanova/sentiment140>
- [2] U. Kayaardi, “Twitter Toxic Tweets,” Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/umitka/twitter-toxic-tweets>
- [3] K. Khudiakova, “Temporal and Behavioral Patterns of Toxicity and Sentiment in Twitter – Source Code,” GitHub. [Online]. Available: https://github.com/Kskksksksksksks/toxicity_analysis